WEEK 3 – SELLER & INVENTORY MANAGEMENT SYSTEM

1. Seller & Inventory Table Design

Seller Table:
Seller_ID – INT – Primary Key
Seller_Name – VARCHAR(100) – NOT NULL
Email – VARCHAR(100) – NOT NULL, UNIQUE
Phone – VARCHAR(15) – NOT NULL, UNIQUE
Address – VARCHAR(200) – NOT NULL

Inventory Table:
Inventory_ID – INT – Primary Key
Product_ID – INT – Foreign Key, NOT NULL
Seller_ID – INT – Foreign Key, NOT NULL
Stock_Quantity – INT – NOT NULL, CHECK
Stock_Status – VARCHAR(20) – NOT NULL, CHECK
Last_Updated – DATE – NOT NULL

2. Updated ER Diagram

SELLER → INVENTORY → PRODUCT

Seller_ID connects Seller and Inventory. Product_ID connects Product and Inventory. One seller can supply many products. A product can have inventory information.

3. Sample Seller & Inventory Data

Seller Data:
1. Arun Traders – arun@gmail.com – 9876543210 – Chennai
2. Kumar Stores – kumar@gmail.com – 9876543211 – Madurai
3. Sri Electronics – srielectronics@gmail.com – 9876543212 – Coimbatore
4. Ravi Enterprises – ravi@gmail.com – 9876543213 – Trichy
5. Vijay Suppliers – vijay@gmail.com – 9876543214 – Salem

Inventory Data:
1. Product 101 – Seller 1 – Stock 25 – Available
2. Product 102 – Seller 1 – Stock 15 – Available
3. Product 103 – Seller 2 – Stock 5 – Available
4. Product 104 – Seller 2 – Stock 0 – Out of Stock
5. Product 105 – Seller 3 – Stock 12 – Available
6. Product 106 – Seller 3 – Stock 8 – Available
7. Product 107 – Seller 4 – Stock 0 – Out of Stock
8. Product 108 – Seller 4 – Stock 30 – Available
9. Product 109 – Seller 5 – Stock 7 – Available

10. Product 110 – Seller 5 – Stock 20 – Available

4. Inventory Analysis Report

Total inventory records: 10
Available products: 8
Out-of-stock products: 2
Highest stock quantity: 30
Lowest stock quantity: 0
Products with stock below 10: 4
Average inventory quantity: 12.2

Each seller supplies 2 products in the sample data. The system helps manage seller details, product assignments, stock quantities, stock status and inventory updates.

5. Query Operations

The SQL script performs the following operations:
Add seller details; assign products to sellers; display products supplied by each seller; count products supplied by each seller; update seller details; display available products; find out-of-stock products; find products with stock less than 10; update stock quantity; delete discontinued inventory; generate seller-wise product report; generate stock availability report; find total available products; find highest stocked product; and calculate average inventory quantity.

Conclusion

The Seller & Inventory Management System successfully manages seller information, product supply details and inventory status using SQL tables, relationships, constraints and analysis queries.